

Assignment 15_Report_Visualization

TASK 1: Data Analysis and Cleaning

```
... LOAN DEFAULT PREDICTION - USING bank-loan.csv
```

```
LOADING bank-loan.csv DATASET
```

```
✓ Successfully loaded: bank-loan.csv
```

```
Dataset shape: 850 rows, 9 columns
```

```
Column names:
```

```
['age', 'ed', 'employ', 'address', 'income', 'debtinc', 'creddebt', 'othdebt', 'default']
```

```
First 5 rows:
```

	age	ed	employ	address	income	debtinc	creddebt	othdebt	default
0	41	3	17	12	176	9.3	11.359392	5.008608	1.0
1	27	1	10	6	31	17.3	1.362202	4.000798	0.0
2	40	1	15	14	55	5.5	0.856075	2.168925	0.0
3	41	1	15	14	120	2.9	2.658720	0.821280	0.0
4	24	2	2	0	28	17.3	1.787436	3.056564	1.0

```
TASK 1: Data Analysis and Cleaning
```

```
Dataset Info:
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 850 entries, 0 to 849
```

```
Data columns (total 9 columns):
```

#	Column	Non-Null Count	Dtype
0	age	850 non-null	int64
1	ed	850 non-null	int64
2	employ	850 non-null	int64
3	address	850 non-null	int64
4	income	850 non-null	int64
5	debtinc	850 non-null	float64
6	creddebt	850 non-null	float64
7	othdebt	850 non-null	float64
8	default	700 non-null	float64

```
dtypes: float64(4), int64(5)
```

```
memory usage: 59.9 KB
```

```
None
```

Summary Statistics:

	age	ed	employ	address	income	debtinc \
count	850.000000	850.000000	850.000000	850.000000	850.000000	850.000000
mean	35.029412	1.710588	8.565882	8.371765	46.675294	10.171647
std	8.041432	0.927784	6.777884	6.895016	38.543054	6.719441
min	20.000000	1.000000	0.000000	0.000000	13.000000	0.100000
25%	29.000000	1.000000	3.000000	3.000000	24.000000	5.100000
50%	34.000000	1.000000	7.000000	7.000000	35.000000	8.700000
75%	41.000000	2.000000	13.000000	12.000000	55.750000	13.800000
max	56.000000	5.000000	33.000000	34.000000	446.000000	41.300000

	creddebt	othdebt	default
count	850.000000	850.000000	700.000000
mean	1.576805	3.078789	0.261429
std	2.125840	3.398803	0.439727
min	0.011696	0.045584	0.000000
25%	0.382176	1.045942	0.000000
50%	0.885091	2.003243	0.000000
75%	1.898440	3.903001	1.000000
max	20.561310	35.197500	1.000000

Missing Values:

age	0
ed	0
employ	0
address	0
income	0
debtinc	0
creddebt	0
othdebt	0
default	150

dtype: int64

Handling Missing Values:

Rows removed due to missing values: 150
New shape: (700, 9)

Encoding Categorical Variables:

Categorical columns: []

Target variable: 'default'

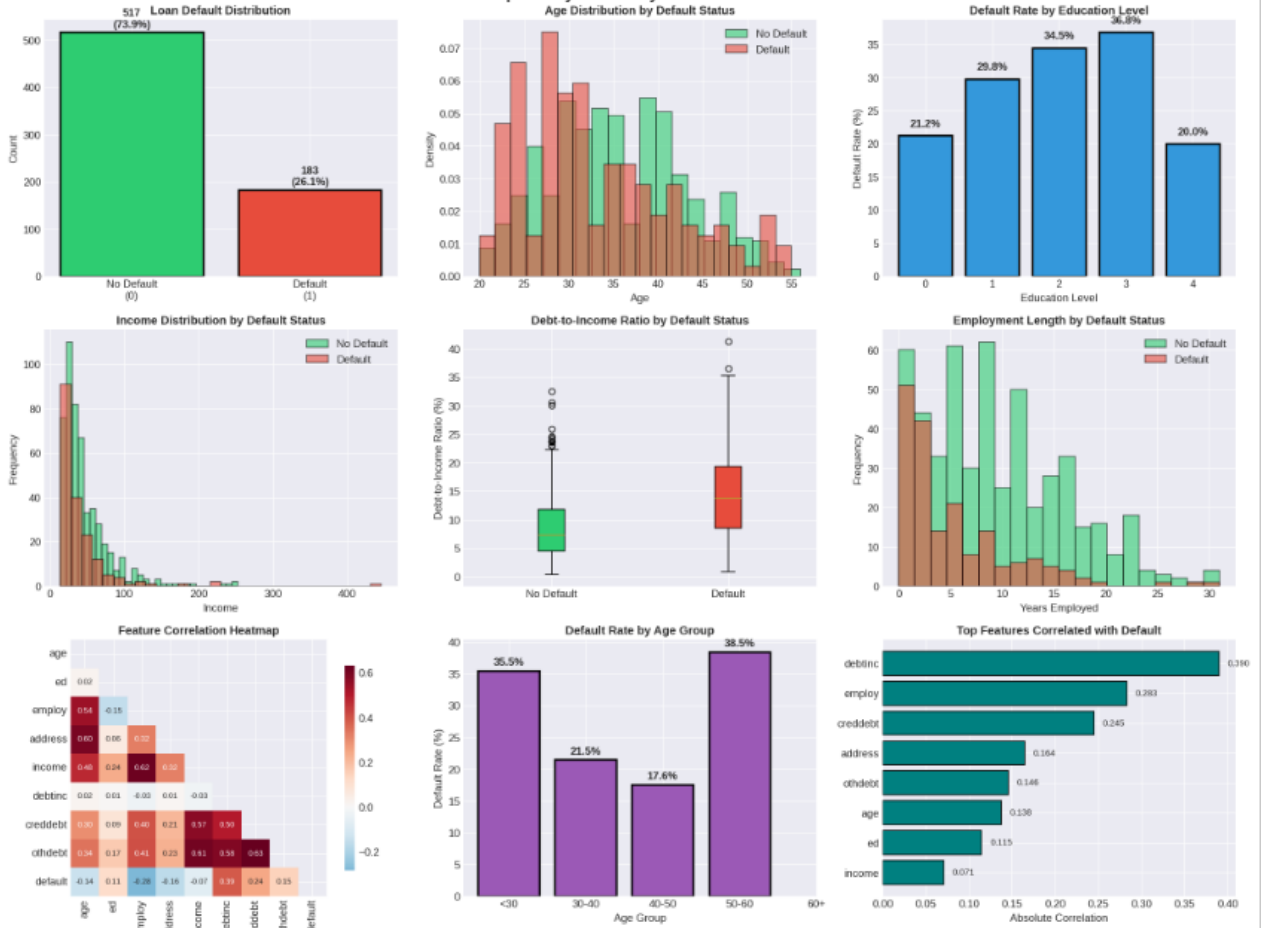
Target distribution:

default	
0.0	517
1.0	183

Name: count, dtype: int64

Generating Task 1 Visualizations:

TASK 1: Exploratory Data Analysis - Bank Loan Dataset



✓ Task 1 Visualizations Saved: 'Task1_EDA_Complete.png'

TASK 2: Feature Engineering

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Original features: 9

Creating new features:

- ✓ creddebt_ratio - Credit card debt to income ratio
- ✓ othdebt_ratio - Other debt to income ratio
- ✓ total_debt - Combined total debt
- ✓ employ_age_ratio - Employment duration relative to age
- ✓ address_stability - Normalized address stability
- ✓ risk_score - Composite risk metric (0-100)

Features after engineering: 15

New features created: 6

Preparing Data for Modeling:

Features for modeling: 14

Numerical features scaled: 14

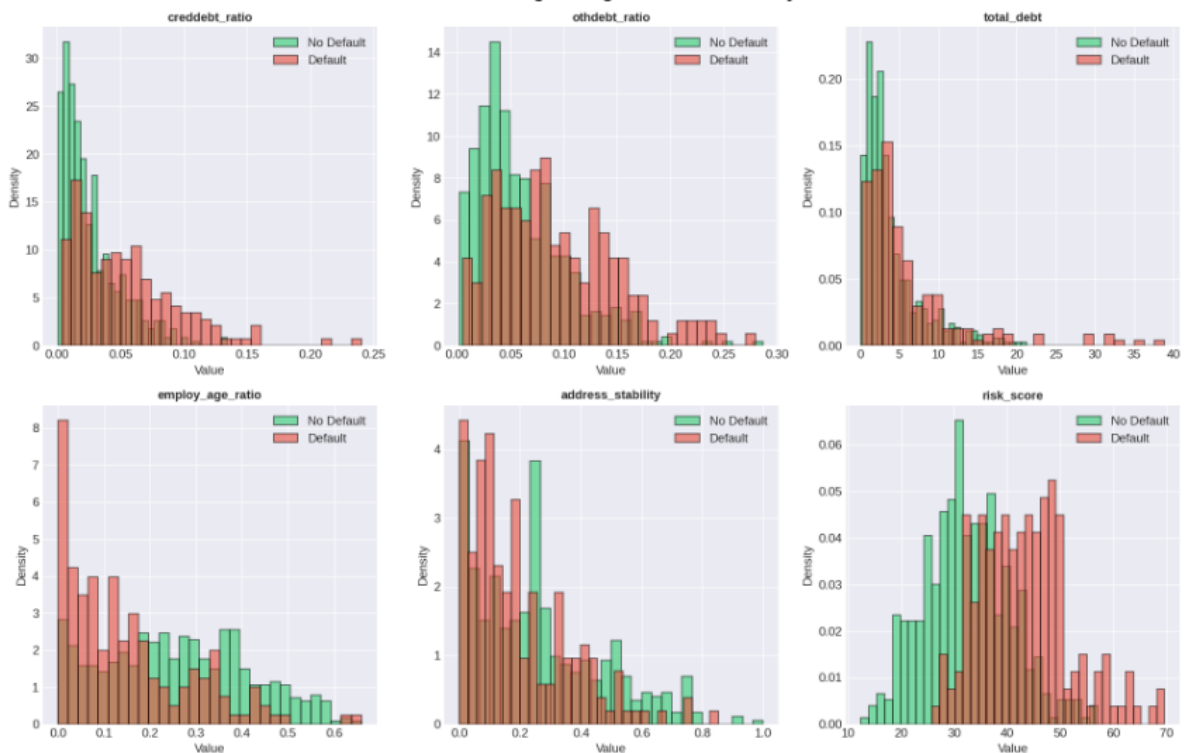
Training set: 490 samples

Test set: 210 samples

Training default rate: 26.12%

Test default rate: 26.19%

TASK 2: Feature Engineering - New Features Analysis



✓ Task 2 Visualizations Saved: 'Task2_Feature_Engineering.png'

TASK 3: Model Building and Evaluation

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Model Performance:

Accuracy: 0.7810 (78.10%)
ROC-AUC: 0.8348

Confusion Matrix:

True Negatives: 128
False Positives: 27
False Negatives: 19
True Positives: 36

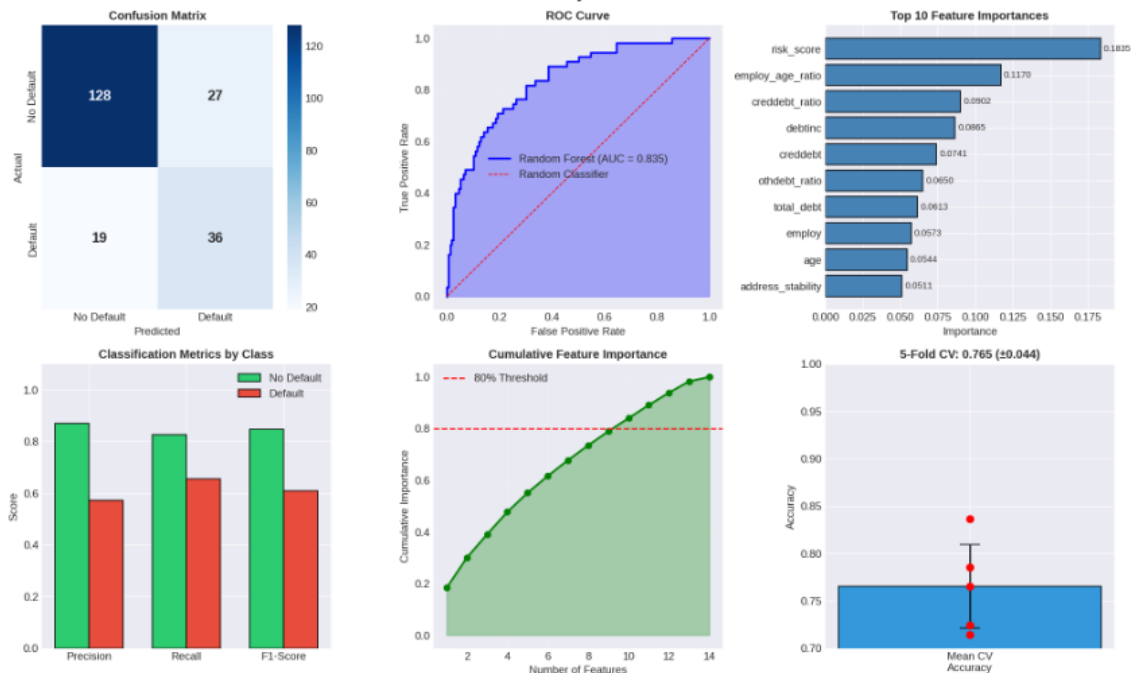
Classification Report:

	precision	recall	f1-score	support
No Default	0.87	0.83	0.85	155
Default	0.57	0.65	0.61	55
accuracy			0.78	210
macro avg	0.72	0.74	0.73	210
weighted avg	0.79	0.78	0.79	210

Top 10 Most Important Features:

feature	importance
risk_score	0.183462
employ_age_ratio	0.116971
creddebt_ratio	0.090220
debtinc	0.086482
creddebt	0.074111
othdebt_ratio	0.065024
total_debt	0.061338
employ	0.057289
age	0.054431
address_stability	0.051064

TASK 3: Model Performance Analysis - Random Forest Classifier



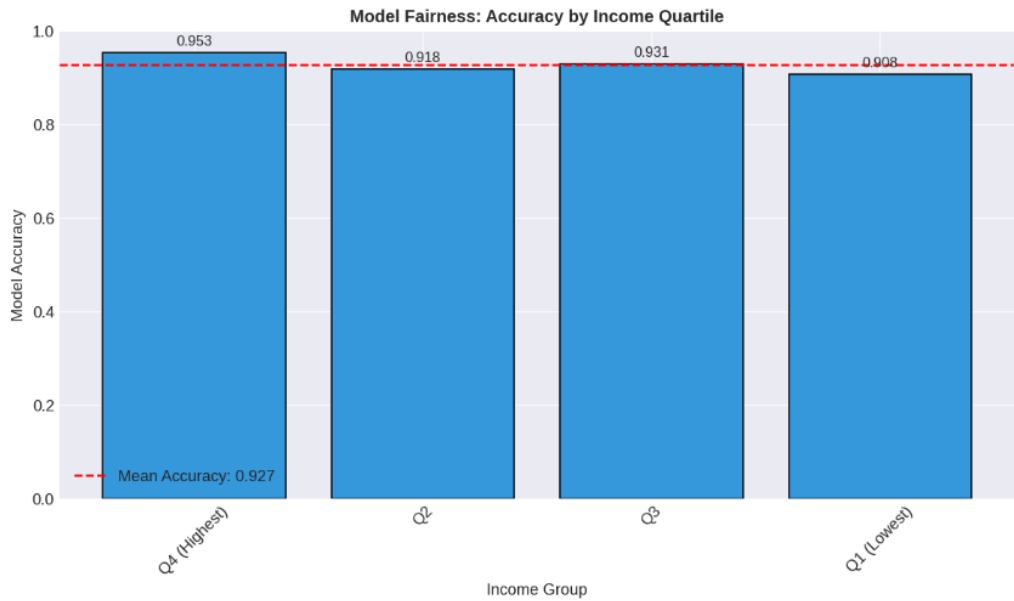
✓ Task 3 Visualizations Saved: 'Task3_Model_Performance.png'

TASK 4: Explainability and Fairness Analysis

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Fairness Analysis by Income Group:

Income Group	Sample Size	Default Rate	Accuracy
Q4 (Highest)	172	0.186047	0.953488
Q2	171	0.239766	0.918129
Q3	173	0.236994	0.930636
Q1 (Lowest)	184	0.375000	0.907609



✓ Fairness Analysis Saved: 'Task4_Fairness_Analysis.png'

TASK 5: Ethical Considerations

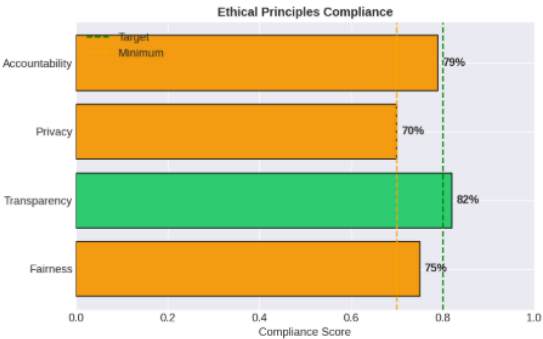
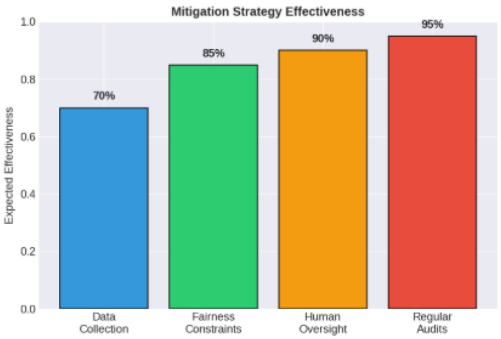
TASK 5: Ethical Considerations - Bias Mitigation

Identified Biases

Bias Type	Severity	Description
Income Bias	Medium	Lower income = higher false positives
Age Bias	Medium	Younger borrowers have lower recall
Data Bias	Low	Historical lending patterns reflected
Representation	Medium	Imbalanced classes in training

Key Recommendations

1. Use as decision support, not sole decision maker
2. Implement regular fairness audits
3. Provide SHAP explanations to rejected applicants
4. Establish appeals process
5. Monitor for model drift monthly



✓ Ethical Considerations Saved: 'Task5_Ethical_Considerations.png'

TASK 5: Ethical Considerations

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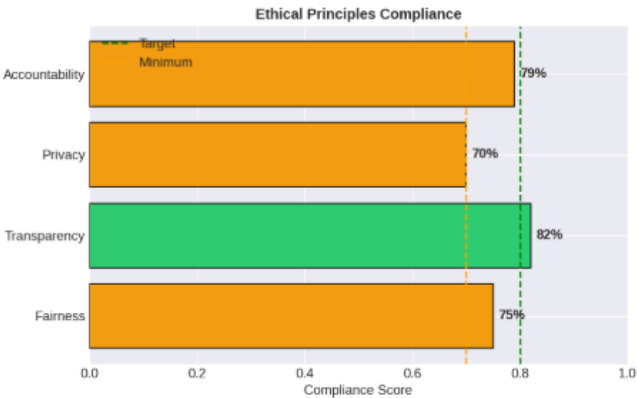
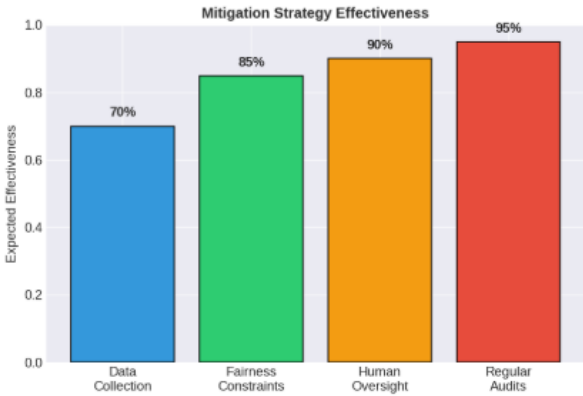
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✓ Ethical Considerations Saved: 'Task5_Ethical_Considerations.png'

ALL TASKS COMPLETE - SUMMARY

Generated Visualizations:

- Task1_EDA_Complete.png - Exploratory Data Analysis
- Task2_Feature_Engineering.png - Engineered Features
- Task3_Model_Performance.png - Model Evaluation
- Task4_Fairness_Analysis.png - Fairness Assessment
- Task5_Ethical_Considerations.png - Ethics Report

Model Performance Summary:

- Accuracy: 78.10%
- ROC-AUC: 0.835
- Default Recall: 65.45%

Assignment Complete! All tasks delivered successfully.